



Snow Mountain Math Adventure

A 5-Day Fun Learning Plan · Year 3 AMC Prep

For: Chloe & a friend (two kids learning together)

Where: On the ski trip — cable car, lodge, snack breaks

Time: 15–20 minutes a day, anytime

Idea: Two kids together = friendly competition + teamwork = more fun than solo

Difficulty: Easy → Hard, climbing over 5 days



How to Play (every day)

- 🏆 **Points:** +1 snowflake ❄️ for each correct answer. No penalty for wrong answers (just like the real AMC!)
- 🐾 **Two modes:**
 - **Battle mode** — first to answer correctly wins the point
 - **Team mode** — discuss together; if right, both get a point
- 🎁 **Reward:** Collect enough ❄️ each day to unlock a treat (hot chocolate, one more run...)
- 🗣️ **Say it out loud:** Whoever answers explains "how I thought about it" — clear explanation earns +1 ❄️



Day 1 — Numbers on the Mountain

Warm-up



Topic: Addition, subtraction & counting

🎮 **Warm-up game "Count the Cable Cars":** On the lift, count together how many towers and pine trees you pass.

1. There are 8 skiers on the slope. 5 more arrive. How many now?
2. You have 12 snowballs and throw 4. How many are left?
3. Each cable car seats 4 people. How many in 3 cars?
4. You start at 9:00 and ski for 2 hours. What time is it?

Answers: 1) 13 2) 8 3) 12 (4×3) 4) 11:00

Day 2 — The Shape of a Snowflake

Geometry



Topic: Shapes, edges & angles, symmetry

Warm-up game "Find Symmetry": Real snowflakes have 6-fold symmetry! Look at gloves and cabin windows — which shapes look the same on both sides when folded?

1. A snowflake has 6 points. How many points do 3 snowflakes have?
2. A triangle has 3 sides and a square has 4 sides. How many sides altogether?
3. Which shape is exactly the same on both sides when folded?  circle ·  square · letter F
4. A cabin window is a rectangle, 5 squares long and 3 squares wide. What is the perimeter (in squares)?

Answers: 1) 18 (6×3) 2) 7 3) circle & square (F is not symmetric) 4) 16 ($(5+3) \times 2$)

Day 3 — Shopping at the Ski Resort

Money



Topic: Money & word problems

Warm-up game "Shopkeeper": One is the shopkeeper, one is the customer — use real coins to practise buying and giving change.

1. A hot chocolate is \$4. Both of you buy one. How much altogether?
2. Chloe has \$10 and buys a \$6.50 sandwich. How much change?
3. A sled rents for \$5 per hour. How much for 3 hours?
4. Gloves \$8, hat \$7, scarf \$5. With \$20, which TWO items can you buy? (List all possibilities.)

Answers: 1) \$8 2) \$3.50 3) \$15 4) gloves+hat (\$15), gloves+scarf (\$13), hat+scarf (\$12) — all three work!

Day 4 — The Slope of Patterns

Logic



Topic: Number patterns & reasoning

Warm-up game "Who's Fastest": Three friends race — give clues and guess the finishing order.

1. Slope flag colours: red, blue, red, blue, red, ____ — what colour next?
2. Number pattern: 5, 10, 15, 20, ____ — what comes next?
3. Anna is faster than Chloe. Chloe is faster than Lily. Who is slowest?
4. Day 1: 2 runs, Day 2: 4 runs, Day 3: 6 runs. At this rate, how many runs on Day 5?

Answers: 1) blue 2) 25 3) Lily 4) 10 runs (add 2 each day: 2,4,6,8,10)

Grand Final



Topic: Mixed problems, counting & multi-step

Final day: Team up and clear all the levels! Get everything right to unlock the title "**Snow Mountain Math Star**"

1. There are red, blue and green hats. Each of you wears one (not the same colour). How many combinations?
2. The lift up takes 15 min and skiing down takes 10 min. How long is one round trip? And 3 round trips?
3. Each snowflake has 6 points. How many snowflakes to reach 30 points?
4. Fold a paper in half 2 times, then unfold — how many small squares? What about 3 folds?

Answers: 1) 6 (red-blue, red-green, blue-red, blue-green, green-red, green-blue) 2) one trip = 25 min; 3 trips = 75 min
3) 5 ($30 \div 6$) 4) 2 folds = 4 squares; 3 folds = 8 squares (doubles each fold)



5-Day Score Sheet

Day	Theme	Chloe ❄️	Friend ❄️
Day 1	Numbers warm-up		
Day 2	Snowflake shapes		
Day 3	Shopping		
Day 4	Patterns		
Day 5	Big challenge		
Total			



Tips for the Grown-up

- **No rush:** Stop when they're tired or want to play — math is the seasoning, not the main course.
- **Encourage explaining:** Explaining it right matters more than getting it right — real understanding shows in the words.
- **Different paces are normal:** Team mode works better than Battle mode when the two kids differ in ability.
- **Hide math in real life:** Cable car numbers, temperature, time, prices... questions are everywhere.